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Thermodynamic modeling of the Ge–Sc system supported by key experiments and first-principles calculation

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ABSTRACT

The Ge–Sc system is assessed by means of CALPHAD (CALculation of PHAse Diagram) approach, supplemented with decisive experiments and enthalpies of formation computed via first-principles calculations. Ten Ge–Sc alloys are prepared by arc melting the pure elements. The annealed samples are analyzed by means of X-ray diffraction, scanning electron microscopy with energy-dispersive X-ray spectrometry, electronic probe microanalysis and differential thermal analysis. Five compounds ($ScGe_2$, ScGe, $Sc_{11}Ge_{10}$, Sc_5Ge_4 and Sc_5Ge_3) are observed in the equilibrium condition, and their enthalpies of formation at 0 K are computed via first-principles calculations to supply the necessary thermodynamic data for the modeling. A set of self-consistent thermodynamic parameters for the Ge–Sc system is finally obtained. The calculated results show a good agreement with the experimental data.

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1. Introduction

Scandium is a powerful dispersoid strengthener, grain refiner and recrystallization inhibitor when added to aluminum alloys [1]. Interaction between scandium and germanium results in the formation of intermetallic compound $ScGe_2$ with a superconducting transition temperature of ~1.3 K [2]. The design of Ge–Sc based novel materials needs information about the phase equilibria and thermodynamic properties in the Ge–Sc system. Meanwhile, the thermodynamic description for the Ge–Sc binary system is also a continuing effort of our previous attempts [3–7] to establish a thermodynamic database for multi-component Al alloys. However, to the best of our knowledge, so far there is no thermodynamic modeling for this binary system.

A thorough thermodynamic assessment of the Ge–Sc system is needed in order to provide a reliable basis for thermodynamic extrapolations and calculations in related ternary and higher-order systems. The purpose of the present work is devoted to providing a self-consistent set of thermodynamic parameters for this binary system by means of a hybrid approach of CALPHAD, key experiments and first-principles calculations.

2. Evaluation of experimental data

The Ge–Sc binary system was critically reviewed by Gokhale and Abbschian [8], who considered the previous experimental results [9]. Most of the results from [9] were accepted by Gokhale and Abbschian [8] assessing the Sc–Ge phase diagram, except for changing the melting temperatures of Sc and Ge from 1823 K and 1208 K to 1814 K and 1211.4 K, respectively, to conform with the currently accepted values. The present assessment not only takes into account the experimental data by Eremenko et al. [9], the Ge–Sc phase diagram assessed by Gokhale and Abbschian [8] and the thermodynamic data [10–13], but also the present new experimental results. The experimental phase diagram data and thermodynamic data are summarized in Table 1, and concise illustration is given as follows.

2.1. Phase diagram data

The major contribution to the Ge–Sc system was due to Eremenko et al. [9]. By means of X-ray diffraction (XRD), metallographic, and differential thermal analysis (DTA), the entire phase diagram was constructed and five compounds $ScGe_2$, ScGe, $Sc_{11}Ge_{10}$, Sc_5Ge_4 and Sc_5Ge_3 were observed. The melting point of Sc_5Ge_3 was determined at 2338 K with a homogeneity range of 2 at.% Ge, whereas the other phases with negligible homogeneities are formed through peritectic reactions, that occur at decreasing temperatures with increasing Ge content. In addition, there are

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Table 1
Summary of the experimental data in the Ge–Sc system.

Type of data	Ref.	Experimental technique[a]	Mode[b]
Phase diagram from 0 to 100 at.% Ge	[9]	XRD, MM, DTA	+
Solubility of Ge in Sc	Present work	XRD, EPMA	+
The homogeneity range of Sc_5Ge_3	Present work	XRD, EPMA	+
Phase diagram from 0 to 16 at.% Ge	Present work	XRD, SEM/EDX, DTA	+
Phase diagram from 60 to 100 at.% Ge	Present work	SEM/EDX, XRD, DTA	+
Enthalpy of mixing of liquid	[10]	Calorimetry	+
Heat capacity of Sc_5Ge_3 from 12 to 300 K	[11]	Adiabatic calorimetry	−
Enthalpy of formation of Sc_5Ge_3	[12]	KHTCR	+
Enthalpies of formation	[13]	emf	−
	Present work	FP	+

[a] MM = micrographic method; XRD = X-ray diffraction; SEM = scanning electron microscopy; EDX = energy dispersive X-ray analysis; DTA = differential thermal analysis; EPMA = electronic-probe microanalysis; emf = electron motive force; FP = first-principles; KHTCR = Kleppa-type high temperature calorimetry.
[b] Indicates whether the data are used or not in the parameter optimization: + used; − not used.

two eutectic reactions $L = (\alpha Sc) + Sc_5Ge_3$ and $L = ScGe_2 + (Ge)$ at 1547 and 1165 K with the eutectic compositions of 12 and 85 at.% Ge, respectively. Meanwhile, the liquidus at the Ge-rich corner was also determined by Eremenko et al. [9].

The crystal structures and lattice parameters for the Sc_5Ge_3, ScGe, and $ScGe_2$ phases were first determined by Arbuckle and Parthe [14], Parthe et al. [15], and Schob and Parthe [16], respectively. Eremenko et al. [9] determined the crystal structures and lattice parameters for the Sc_5Ge_4 and $Sc_{11}Ge_{10}$ phases. The lattice parameters mentioned above are confirmed with our experimental results, as shown later, and used in the present first-principles calculation.

2.2. Thermodynamic data

By means of a high-temperature isoperibolic calorimeter, the enthalpy of mixing in the liquid phase was determined at 1973 K by Beloborodova [10] with the composition range from 60 to 100 at.% Ge, which was used in the present optimization. The low-temperature heat capacity and enthalpy of formation for the Sc_5Ge_3 phase were measured by Belokurov et al. [11] employing adiabatic calorimetry and Jung and Kleppa [12] using high-temperature calorimetry, while Lukashenko et al. [13] obtained the enthalpies of formation for all the compounds in the Ge–Sc system through electromotive force (emf) measurement of high temperature galvanic cells. However, the reported enthalpies of formation [12,13] for the Sc_5Ge_3 phase exhibit large discrepancies. As a result, the first-principles calculation is applied in the present work to select which one is preferable.

3. Experimental procedure

Ten Ge–Sc binary alloys, each with a weight of about 1 g, were prepared by melting the mixture of pure Ge (99.9 wt%) and Sc (99.9 wt%) according to the designated compositions in an arc-melting furnace (WKDHL-1, Opto-electronics Co. Ltd., Beijing, China) with a non-consumable *W* electrode under high pure Ar atmosphere. The buttons were remelted five times to improve their homogeneities. The as-cast alloys were not subjected to chemical analysis since the weight losses were less than 0.5 wt% for all of the prepared alloys. The as-cast alloys were then sealed in evacuated quartz capsules under vacuum (4×10^{-4} bar) and annealed in an L4514-type diffusion furnace (Qingdao Instrument & Equipment Co. Ltd., China) at specified temperatures for 40 days, as shown in Table 2, followed by quenching in water.

The XRD powder measurement of the annealed alloys was performed using Cu-K_α radiation on a Rigaku D-max/2550 VB+ X-ray diffractometer at 40 kV and 300 mA. The samples, which were characterized as two-phase by XRD, were further examined using scanning electron microscopy (SEM) equipped with energy dispersive X-ray analysis (EDX) (JSM-5600LV, JEOL, Japan). Alloys 1–9 were then subjected to DTA (DSC404C, Netzsch, Germany) measurements. The measurement was performed between room temperature and 1723 K with heating and cooling rates of 5 K min^{-1} in flowing Ar. The accuracy of temperature measurement using a Pt–Pt/Rh thermocouple was estimated to be ±2 K. The liquidus was determined from the peak temperature of the final thermal effect on heating, and the invariant temperatures from the onset of the other thermal effects during the heating step.

In order to determine the homogeneity range of Sc_5Ge_3 and the solubility of Ge in Sc at 1323 and 1223 K, alloys 3 and 10, which are located in two-phase regions of $(\alpha Sc) + Sc_5Ge_3$ and $Sc_5Ge_3 + Sc_5Ge_4$, respectively, were analyzed by electronic probe microanalysis (EPMA) (JXA-8800R, Electron Optics. Ltd., Tokyo, Japan) under the operation condition of 20 kV voltage, 20 mA current and a 40° take-off angle.

4. First-principles calculations for enthalpy of formation

In the present work, first-principles calculations based on density functional theory (DFT) [17] within generalized gradient approximation (GGA) along with projector augmented-wave (PAW) method [18] are employed to obtain $\Delta_f H$ for the compounds Sc_5Ge_3, Sc_5Ge_4, $Sc_{11}Ge_{10}$, ScGe, and $ScGe_2$, as implemented in the Vienna ab initio simulation package (VASP) [19,20]. The GGA proposed by Perdew et al. [21] is used in the calculations. Brillouin zone integrations are performed using a Monkhorst–Pack mesh [22], and the sampling of *k*-point mesh is $6 \times 6 \times 7$ for Sc_5Ge_3, $6 \times 3 \times 6$ for Sc_5Ge_4, $4 \times 4 \times 3$ for $Sc_{11}Ge_{10}$, $17 \times 7 \times 19$ for GeSc, and $17 \times 4 \times 17$ for $ScGe_2$. The energy cut off at 400 eV is sufficient to ensure the total energy differences are less than 1 meV/atom. The structure is fully relaxed using the Methfessel–Paxton method [23], and a final self-consistent static calculation via the tetrahedron method with Blochl corrections [24] is performed.

The enthalpy of formation of a compound can be defined as the difference in total energy of the compound and the energies of its constituent elements in their stable states:

$$\Delta_f H = E_{\text{total}}^{Sc_xGe_{1-x}} - x \cdot E_{\text{total}}^{\text{Sc-HCP}} - (1-x) \cdot E_{\text{total}}^{\text{Ge-Diamond}} \quad (1)$$

where $E_{\text{total}}^{Sc_xGe_{1-x}}$ is the total energy of the compound Sc_xGe_{1-x}. $E_{\text{total}}^{\text{Sc-HCP}}$ and $E_{\text{total}}^{\text{Ge-Diamond}}$ represent the total energies of Sc and Ge in their stable structures, respectively. Since the influence of pressure on the condensed phases was ignored and the energy was calculated at 0 K without any entropic contributions, the calculated energy of formation was taken to be equivalent to the enthalpy of formation at 298 K. The calculated enthalpy of formation for Sc_5Ge_3 (−84.7 kJ/mol-atoms) is close to the result of

Table 2
Summary of the phase and DTA signal for the Ge–Sc alloys annealed at specified temperatures for 40 days.

Alloy		Annealing temperature (K)	Phase[a]	DTA signal (K)[b]
No.	Composition (at.% Ge)			
1	5	1323	$(\alpha Sc) + Sc_5Ge_3$	1579.2, 1611
2	12	1323	$(\alpha Sc) + Sc_5Ge_3$	1546.9, 1576.2
3[c]	16	1323	$(\alpha Sc) + Sc_5Ge_3$	1543
		1223		
4	60	1123	$ScGe + ScGe_2$	1389, 1466
5	66.67	1123	$ScGe_2$	1389, 1466, 1475.1
6	67	1123	$ScGe_2 + (Ge)$	1167, 1390
7	70	1123	$ScGe_2 + (Ge)$	1167, 1370.6
8	75	1123	$ScGe_2 + (Ge)$	1170.4, 1317.6
9	90	1123	$ScGe_2 + (Ge)$	1172.2, 1185.6
10[c]	41	1323	$ScGe_2 + (Ge)$	
		1223		

[a] Identified with XRD and SEM/EDX.
[b] Obtained from DTA measurement with a heating rate of 5 K min^{-1}.
[c] The samples were also analyzed by EPMA to clarify the homogeneity rang of Sc_5Ge_3 and the solubility of Ge in Sc.

Table 3
Summary of the phases, their compositions and lattice parameters for the Ge–Sc alloys.

Phase	Space group	Lattice parameters (nm)			Composition (at.% Ge)	Refs.
		a	*b*	*c*		
(αSc)	$P6_3/mmc$	0.33088		0.52680	2.7(max)	[29,30]
		0.33107(7)		0.52693(9)	1.46[a], 2.41[b]	Exp.
		0.33082		0.51769		F.P.
(βSc)	$Im\bar{3}m$	0.3730			2(max)	[31]
Sc_5Ge_3	$P6_3/mcm$	0.7939		0.5883	36.5–38.5	[9,14]
		0.7932(1)		0.5898(5)	35.9–37.8[a]	Exp.
					35.6–38[b]	Exp.
		0.7972		0.5905		F.P.
Sc_5Ge_4	$Pnma$	0.7269(2)	1.3980(3)	0.7152(8)	44.5	[9]
		0.7172	1.3807	0.7251		F.P.
$Sc_{11}Ge_{10}$	$I4/mmm$	1.012(2)		1.554(2)	47.62	[9]
		1.032		1.530		F.P.
ScGe	$Cmcm$	0.4007(4)	1.0060(1)	0.3762(4)	50	[9,15,16]
		0.4025	1.0133	0.3825		F.P.
$ScGe_2$	$Cmcm$	0.388(8)	1.487(3)	0.379(3)	66.67	[9,16]
		0.388(8)	1.485(7)	0.379(5)	66.67	Exp.
		0.388	1.487	0.379		F.P.
(Ge)	$Fd\bar{3}m$	0.5657			100	[30]
		0.5620(3)			100	Exp.
		0.5620				F.P.

[a] The phase composition at 1223 K measured by EMPA.
[b] The phase composition at 1323 K measured by EMPA. EXP. = Present Experimental results; F.P. = Present First-Principles results.

Jung and Kleppa [12] (-93.4 ± 3.9 kJ/mol-atoms). However, when compared with the result of Lukashenko et al. [13], a deviation of more than 10 kJ/mol-atoms appears for all compounds. In this situation, the first-principles data are considered superior since unrealistic entropies may be caused by experimental values, more detail of which has been discussed in [25]. Therefore, the result of Jung and Kleppa [12] and the presently calculated results are accepted in the thermodynamic optimization.

5. Thermodynamic model

5.1. Unary phases

The Gibbs energy function ${}^0G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}$ for the pure element i (i = Sc, Ge) is expressed as:

$$ {}^0G_i^{\varphi}(T) = a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1} + f \cdot T^3 + g \cdot T^7 + h \cdot T^{-9} \quad (2) $$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its stable element reference (SER) state, and T is the absolute temperature. In the present modeling, the Gibbs energies as a function of temperature are taken from the SGTE compilation by Dinsdale [26].

5.2. Solution phases

The Gibbs energy for the solution phase (liquid, (αSc) and (βSc)) is described by the Redlich–Kister polynomial for the excess Gibbs energy [27]:

$$ G_m^L - H^{SER} = x_{Ge} \cdot {}^0G_{Ge}^L + x_{Sc} \cdot {}^0G_{Sc}^L + R \cdot T \cdot [x_{Ge} \cdot \ln x_{Ge} + x_{Sc} \cdot \ln x_{Sc}] + x_{Ge} \cdot x_{Sc} \cdot [a_0 + b_0 \cdot T + (x_{Ge} - x_{Sc}) \cdot (a_1 + b_1 \cdot T) + \cdots] \quad (3) $$

in which H^{SER} denotes $x_{Ge} \cdot H_{Ge}^{SER} + x_{Sc} \cdot H_{Sc}^{SER}$, R is the gas constant, and x_{Ge} and x_{Sc} represent the mole fractions of Ge and Sc, respectively.

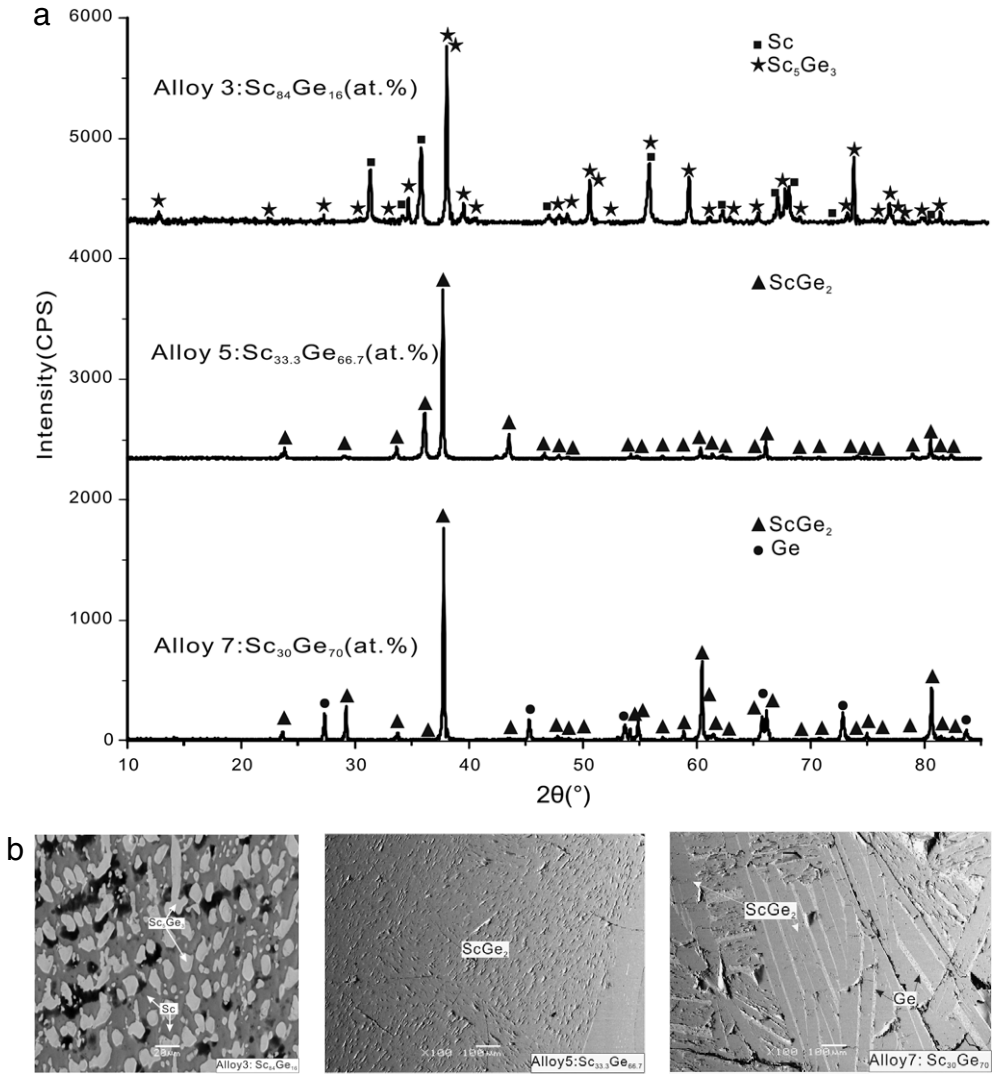

Fig. 1. (a) The XRD patterns of alloys 3 ($Sc_{84}Ge_{16}$), 5 ($Sc_{33.3}Ge_{66.7}$) and 7 ($Sc_{30}Ge_{70}$) annealed at 850 °C for 40 days; (b) the backscattered electron image (BEI) of alloy 3 ($Sc_{84}Ge_{16}$), 5 ($Sc_{33.3}Ge_{66.7}$) and 7 ($Sc_{30}Ge_{70}$).

The interaction coefficients a_j and b_j ($j = 0, 1$) are to be evaluated in the optimization process.

5.3. Intermetallic compounds

The compounds Sc_5Ge_4, $Sc_{11}Ge_{10}$, ScGe and $ScGe_2$ are modeled as stoichiometric phases. The Gibbs energy of Sc_xGe_y phase ($Sc_xGe_y = Sc_5Ge_4$, $Sc_{11}Ge_{10}$, ScGe or $ScGe_2$) per mole-formula is given by the following expression:

$$G_m^{Sc_xGe_y} - x \cdot H_{Sc}^{SER} - y \cdot H_{Ge}^{SER} = A + B \cdot T + x \cdot {}^0G_{Sc}^{HCP_A3} + y \cdot {}^0G_{Ge}^{Diamond} \quad (4)$$

where A and B are the parameters to be evaluated in the course of optimization.

In view of reported homogeneity ranges [9] and present EPMA results, the Sc_5Ge_3 phase is described with the two-sublattice model (**Sc**, Ge)$_5$(Sc, **Ge**)$_3$. The bold faces mean the normal atoms (i.e., major species) in the sublattices. The Gibbs energy of Sc_5Ge_3 phase per mole-formula can thus be expressed as follows:

$$\begin{aligned} {}^0G_m^{Sc_5Ge_3} &= y'_{Sc} \cdot y''_{Ge} \cdot {}^0G_{Sc:Ge}^{Sc_5Ge_3} \\ &+ y'_{Sc} \cdot y''_{Sc} \cdot {}^0G_{Sc:Sc}^{Sc_5Ge_3} + y'_{Ge} \cdot y''_{Sc} \cdot {}^0G_{Ge:Sc}^{Sc_5Ge_3} \\ &+ y'_{Ge} \cdot y''_{Ge} \cdot {}^0G_{Ge:Ge}^{Sc_5Ge_3} \\ &+ 5 \cdot R \cdot T \cdot (y'_{Sc} \cdot \ln y'_{Sc} + y'_{Ge} \cdot \ln y'_{Ge}) \\ &+ 3 \cdot R \cdot T \cdot (y''_{Sc} \cdot \ln y''_{Sc} + y''_{Ge} \cdot \ln y''_{Ge}) \\ &+ y'_{Sc} \cdot y''_{Sc} \cdot y''_{Ge} \cdot {}^0L_{Sc:Sc,Ge}^{Sc_5Ge_3} \\ &+ y'_{Ge} \cdot y''_{Sc} \cdot y''_{Ge} \cdot {}^0L_{Ge:Sc,Ge}^{Sc_5Ge_3} \\ &+ y'_{Sc} \cdot y'_{Ge} \cdot y''_{Ge} \cdot {}^0L_{Sc,Ge:Ge}^{Sc_5Ge_3} \\ &+ y'_{Sc} \cdot y'_{Ge} \cdot y''_{Sc} \cdot {}^0L_{Sc,Ge:Sc}^{Sc_5Ge_3} + \cdots \end{aligned} \quad (5)$$

where the parameter y_i^j is the site fraction of the species i (i = Sc or Ge) on the sublattice j. The superscripts $'$ and $''$ denote the first and second sublattices of the present model, respectively. ${}^0G_{Sc:Ge}^{Sc_5Ge_3}$ is the Gibbs energy of ideal compound Sc_5Ge_3 and can be expressed as an equation similar to Eq. (4). The first-principles calculated positive values of 13.444 k J /mol-atoms and 34.243 k J/mol-atoms relative pure elements are assigned to ${}^0G_{Sc:Sc}^{Sc_5Ge_3}$ and ${}^0G_{Ge:Ge}^{Sc_5Ge_3}$, respectively. The parameter ${}^0G_{Ge:Sc}^{Sc_5Ge_3}$ relates to the antistructure defects in both sublattices, and it can be estimated by the equation ${}^0G_{Ge:Sc}^{Sc_5Ge_3} = {}^0G_{Sc:Sc}^{Sc_5Ge_3} + {}^0G_{Ge:Ge}^{Sc_5Ge_3} - {}^0G_{Sc:Ge}^{Sc_5Ge_3}$. In addition, the interaction parameter ${}^0L_{Sc:Sc,Ge}^{Sc_5Ge_3}$ is treated

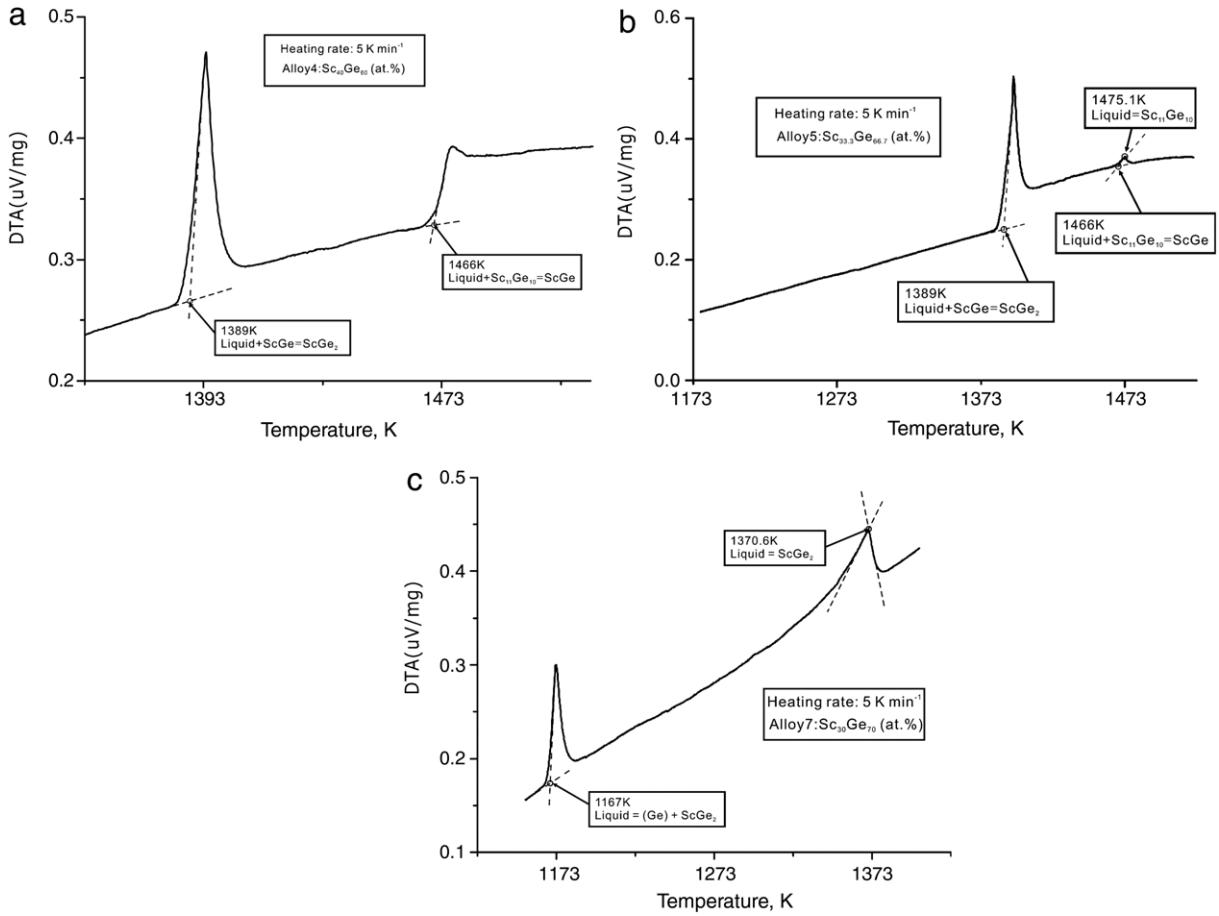

Fig. 2. (a) DTA curve of alloy 4 ($Sc_{40}Ge_{60}$) with a heating rate of 5 K min^{-1}; (b) DTA curve of alloy 5 ($Sc_{33.3}Ge_{66.7}$) with a heating rate of 5 K min^{-1}; (c) DTA curve of alloy 7 ($Sc_{30}Ge_{70}$) with a heating rate of 5 K min^{-1}.

as $C + D \cdot T$, where C and D are parameters to be determined in this assessment.

6. Results and discussion

6.1. Experimental results

All experimental results for the phases identified with XRD and SEM, their lattice parameters by XRD and the phase compositions resulting from EPMA measurement are summarized in Table 3. The lattice parameters of identified phases were refined with the Jade 5 software [28] and compared with the literature [9] as well as the present first-principles calculations.

The XRD patterns and backscattered electron image (BEI) of the representative alloys 3 ($Sc_{84}Ge_{16}$), 5 ($Sc_{33.3}Ge_{66.7}$) and 7 ($Sc_{30}Ge_{70}$) are shown in Fig. 1(a) and (b), respectively. As indicated in these figures, alloy 3 is located in the two-phase region of (αSc)+Sc_5Ge_3, and alloy 7 is located in the two-phase region of $ScGe_2$+(Ge). Alloy 5 is in the $ScGe_2$ single phase region.

According to EPMA measurement, the solubility of Ge in (αSc) is 1.46 and 2.41 at.% Ge at 1223 K and 1323 K, respectively. The homogeneity range of Sc_5Ge_3 at 1223 K and 1323 K are detected as 35.9–37.8 at.% Ge and 35.6–38 at.% Ge, respectively.

Table 2 summaries the phase transition temperatures obtained by DTA measurement, which directly confirm the general features of the Ge–Sc phase diagram established by Eremenko et al. [9]. The presently determined two invariant reactions on the Sc-rich side ((βSc) = (αSc) + Liquid at1605 K, Liquid = (αSc) + Sc_5Ge_3 at 1547 K) match well with those reported in the previous work [9]. On the Ge-rich side, three invariant equilibria ($Sc_{11}Ge_{10}$+Liquid = ScGe, ScGe + Liquid = $ScGe_2$ and Liquid = $ScGe_2$ + (Ge)) were detected at 1466 K, 1389 K and 1167 K, respectively, showing a good agreement with previous results [9].

However, the peritectic point of the invariant reaction $Sc_{11}Ge_{10}$ + Liquid = ScGe as well as the liquidus above it represented some differences between the present DTA results and the data in [9]. Fig. 2(a) and (b) are the DTA curve of alloys 4 ($Sc_{40}Ge_{60}$) and 5 ($Sc_{33.3}Ge_{66.7}$), respectively, where two reaction peaks were detected in both alloys. For alloy 4, the first peak represents the peritectic reaction (ScGe + Liquid = $ScGe_2$) at 1389 K, while the second peak indicates another peritectic reaction ($Sc_{11}Ge_{10}$ + Liquid = ScGe) at 1466 K. The above two peak temperatures are also detected in alloy 5. Since alloy 5 is determined to be $ScGe_2$ single phase according to XRD, it is obvious that the reaction (L + $Sc_{11}Ge_{10}$ = ScGe) does not terminate at 62.5 at.% Ge as reported in [9], but extends to above 66.7 at.% Ge. The liquidus for $Sc_{11}Ge_{10}$ alloy reported by Eremenko et al. [9] is thus not reasonable and the temperature of 1475.1 K from the peak maximum of the second thermal effect is assumed as the liquidus point in the present assessment.

Fig. 2(c) gives the DTA curve of alloy 7 ($Sc_{30}Ge_{70}$) with a heating rate of 5 K min^{-1}, the two peaks represent the invariant equilibria (Liquid = $ScGe_2$ + (Ge)) at 1167 K and the temperature of liquidus at this composition (1370.6 K). These experimental data confirm the phase relation in the Ge-rich side and are accepted in the present optimization.

6.2. Thermodynamic assessment

The optimization of the thermodynamic parameters in the Ge–Sc system was performed with the PARROT module of

Table 4
Summary of the thermodynamic parameters in the Ge–Sc system.[a]

Liquid: Model $(Ge, Sc)_1$
$^0L^{Liquid}_{Ge,Sc} = -269889.79 + 14.6392 \cdot T$
$^1L^{Liquid}_{Ge,Sc} = 48459.76 - 19.9950 \cdot T$
$^2L^{Liquid}_{Ge,Sc} = 72086.29$

(αSc): Model $(Ge, Sc)_1(Va)_{0.5}$
$^0L^{(\alpha Sc)}_{Ge,Sc} = -186302.61$

(βSc): Model $(Ge, Sc)_1(Va)_3$
$^0L^{(\beta\beta Sc)}_{Ge,Sc} = -183087.40$

$ScGe_2$:$Sc_{0.33333}Ge_{0.66667}$
$^0G^{ScGe_2}_{Ge:Sc} - 0.33333 \cdot G^{SER}_{Sc} - 0.66667 \cdot G^{SER}_{Ge} = -56894.05 - 3.6909 \cdot T$

ScGe: $Sc_{0.5}Ge_{0.5}$
$^0G^{ScGe}_{Ge:Sc} - 0.5 \cdot G^{SER}_{Sc} - 0.5 \cdot G^{SER}_{Ge} = -82133.92$

$Sc_{11}Ge_{10}$:$Sc_{0.52381}Ge_{0.47619}$
$^0G^{Sc_{11}Ge_{10}}_{Ge:Sc} - 0.52381 \cdot G^{SER}_{Sc} - 0.47619 \cdot G^{SER}_{Ge} = -82870.86 - 1.2751 \cdot T$

Sc_5Ge_4:$Sc_{0.55556}Ge_{0.44444}$
$^0G^{Sc_5Ge_4}_{Ge:Sc} - 0.55556 \cdot G^{SER}_{Sc} - 0.44444 \cdot G^{SER}_{Ge} = -84474.86 - 0.8592 \cdot T$

Sc_5Ge_3:$(Sc, Ge)_{0.625}(Sc, Ge)_{0.375}$
$^0G^{Sc_5Ge_3}_{Sc:Ge} - 0.625 \cdot G^{SER}_{Sc} - 0.375 \cdot G^{SER}_{Ge} = -83831.65$
$^0G^{Sc_5Ge_3}_{Sc:Sc} - G^{SER}_{Sc} = 13444$
$^0G^{Sc_5Ge_3}_{Ge:Ge} - G^{SER}_{Ge} = 34243$
$^0G^{Sc_5Ge_3}_{Ge:Sc} = {}^0G^{Sc_5Ge_3}_{Sc:Sc} + {}^0G^{Sc_5Ge_3}_{Ge:Ge} - {}^0G^{Sc_5Ge_3}_{Sc:Ge}$
$^0L^{Sc_5Ge_3}_{Sc:Ge,Sc} = -1522.86$

[a] Temperature (T) in Kelvin and Gibbs energy in J/mol.

the Thermo-calc software [32]. The step-by-step optimization procedure described by Du et al. [33] was adopted in the present assessment.

The optimization began with the liquid phase. First, the experimental enthalpy of mixing in the liquid phase at Ge-rich side was considered. Second, the intermetallic phases were considered in the optimization. In the case of the compounds, the compound Sc_5Ge_3 was taken into account first. Then the other intermetallic phases were introduced in the modeling one by one. The first-principles computed enthalpy of formation for all compounds and the experimental value of Sc_5Ge_3 measured by Jung and Kleppa [12] were separately used as the starting value for the parameter A in the present modeling. Parameters A and B were adjusted to describe the liquidus and invariant reactions. It turned out that the first-principles result of Sc_5Ge_3 could make a better overall fitting. Thus the literature data [12] was given a small weight during the present assessment. Third, one regular parameter was employed for each of the (αSc) and (βSc) phases in order to account for the solubility data. Finally, the thermodynamic parameters for all the phases were optimized simultaneously. The optimized thermodynamic parameters are listed in Table 4.

The computed Ge–Sc phase diagram using the present set of thermodynamic parameters is shown in Fig. 3, and compared with the literature data [9] and the present experimental results. The calculated invariant equilibria are compared with the experimental ones in Table 5. It can be seen that the present modeling agrees well with the experimental data from Ref. [9]. The difference between the calculated and the measured invariant equilibrium temperatures are less than 5 K. The calculated reaction temperature and peritectic point of invariant equilibrium $Sc_{11}Ge_{10}$ + Liquid = ScGe are 1389 K and 68.1 at.% Ge, which agree well with the present experimental results.

Fig. 4 presents the calculated enthalpies of mixing for liquid at 1973 K compared with the experimental data [10]. It can be seen that the calculation can satisfactorily account for the experimental ones.

Fig. 5 presents the calculated enthalpies of formation from CALPHAD modeling compared with those from the first-principles method and experimental data [12,13]. The present calculation results agree well with the first-principles results.

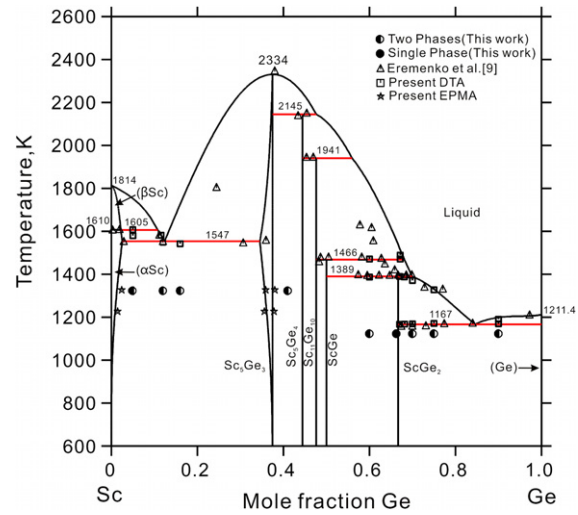

Fig. 3. Calculated Ge–Sc phase diagram with the experimental data from the present work and the literature [9].

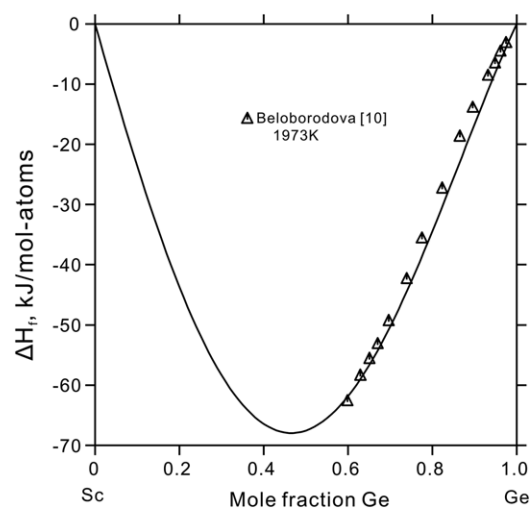

Fig. 4. Calculated enthalpies of mixing for liquid at 1973 K, compared with the experimental data [10]. The reference states are liquid Ge and liquid Sc.

7. Conclusions

- All the experimental phase diagram and thermodynamic data available for the Ge–Sc system have been critically evaluated. Ten alloys were prepared and analyzed by XRD, SEM, EPMA and DTA to supply the thermodynamic modeling. The enthalpies of formation for the binary compounds computed via first-principles calculations are used in the present thermodynamic modeling.
- An optimum set of thermodynamic parameters for the Ge–Sc system was obtained by considering the present results and critically assessed literature data. The comprehensive comparison shows that the calculated phase diagram and thermodynamic properties are in a good agreement with the experimental data and first-principles calculations.

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Table 5
Summary of the invariant equilibria in the Ge–Sc system.

Reaction	Type	T (K)	Composition (at.% Ge)			Ref.	Method
(βSc) = (αSc) + Liquid	Metatectic	1603	~2	~0	~8.9	[8]	Assessed
		1605				[This work]	DTA[a]
		1605	2.21	2.17	10.7	[This work]	Calculated
Liquid = (αSc) + Sc_5Ge_3	Eutectic	1547	12	2.7	36.5	[8]	DTA[a]
		1547				[This work]	DTA[a]
		1547	12.3	2.67	34.53	[This work]	Calculated
Sc_5Ge_3 = Liquid	Congruent	2338	37.5	37.5		[8]	DTA[a]
		2334	37.5	37.5		[This work]	Calculated
Sc_5Ge_3 + Liquid = Sc_5Ge_4	Peritectic	2143	38.5	~45.2	44.45	[8]	DTA[a]
		2145	37.5	47.6	44.45	[This work]	Calculated
Sc_5Ge_4 + Liquid = $Sc_{11}Ge_{10}$	Peritectic	1938	44.45	~49.3	47.62	[8]	DTA[a]
		1941	44.45	55.9	47.62	[This work]	Calculated
$Sc_{11}Ge_{10}$ + Liquid = ScGe	Peritectic	1465	47.62	62.5	50	[8]	DTA[a]
		1466		>66.7		[This work]	DTA[a]
		1466	47.62	67.7	50	[This work]	Calculated
ScGe + Liquid = $ScGe_2$	Peritectic	1389	50	69.9	66.7	[8]	DTA[a]
		1389				[This work]	DTA[a]
		1389	50	69.9	66.7	[This work]	Calculated
Liquid = $ScGe_2$ + (Ge)	Eutectic	1165	85	66.7	100	[8]	DTA[a]
		1167				[This work]	DTA[a]
		1167	85	66.7	100	[This work]	Calculated

[a] DTA: differential thermal analysis.

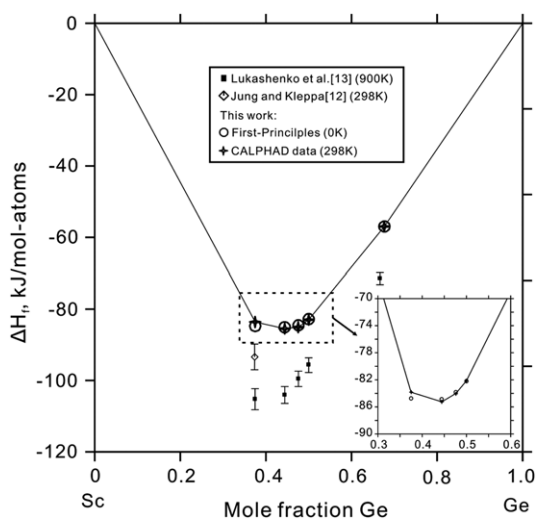

Fig. 5. Calculated standard enthalpies of formation of solid phases at 298.15 K, compared with the first-principles and experimental data [12,13]. The data in Ref. [13] are plotted with error bars and not used in the optimization.

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Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2011.12.010.

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